GMR Institute of Technology
An Autonomous Institute Affiliated to JNTU-GV

GMR Institute of Technology

Department of CSE-AI&ML

Title of the Project : **Smart Paper Checker: An NLP-Based System for Automatic Subjective Answer Evaluation**

Name of the Course : B.Tech. (CSE-AI&ML) Semester: 7th Academic Year: 2025-2026

ABSTRACT:

In the wake of rising online education and the shift toward virtual examinations, manual evaluation of subjective answers has proven to be time-consuming, error-prone, and inefficient. To address this challenge, this project proposes a future-ready system that leverages Natural Language Processing (NLP) to automate the correction of written responses in exams. The envisioned system will use NLP techniques such as tokenization, stop-word removal, stemming, and semantic similarity to compare student answers against reference solutions and assign scores accordingly. Initially, the focus will be on typed subjective answers to simplify processing, with plans to later incorporate Optical Character Recognition (OCR) for handwritten answers. The backend will be developed using Python, integrating libraries like NLTK, while the user interface will be built with web technologies such as Flask and HTML/CSS. The goal of this future implementation is to reduce the workload of educators, ensure unbiased grading, and provide students with instant, transparent feedback. This project aims to support educational institutions by enhancing the scalability and fairness of digital assessment systems. Upon successful completion, the system could serve as a stepping stone toward more intelligent, AI-driven exam correction platforms.

Keywords: *Natural Language Processing (NLP), Tokenization, Stop-Word Removal, Stemming, Optical Character Recognition (OCR), Natural Language Tool Kit(NLTK).*

Base Paper:

Nagarajan, M. K., Subramanian, R. R., Akshaya, S., & Sreya, B. (2022). "Automatic paper corrector using NLP". International Journal of Emerging Technology in Computer Science & Electronics (IJETCSE), 29(3).

Project Supervisor
Dr. P. Satish Kumar

M. Praveen 22341A4232
D. Rohith 22341A4207
T. Swetha 22341A4258
K. Naveen 22341A4216
V. Eswar Prasad 22341A4262